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1.Project Description

1.Context

Academic failure is often identified only after it has occurred. Our goal is to use information available before the exam season to identify students who may be at risk of failing and may require additional support.

2.Problem statement

How can we use such information to identify students who are at risk of failing and determine the main factors associated with it?

3. Project Objectives

1. Early Risk Detection

Identify students who are at risk of failing before the exam season.

2. Identify main Contributing Factors

Determine which academic, behavioral, lifestyle, and other student factors are most correlated with failure.

3. Actionable Support

Provide insights that can help educational stakeholders take timely and targeted actions to support students that are at risk

4.Dataset:

The dataset contains information from several domains. It includes realistic interactions among student habits, socioeconomic backgrounds, study behaviors, and mental wellbeing, offering an ideal sandbox for machine learning modeling, data cleaning, and exploratory analysis.

- **Source:** Kaggle — Mobeen Fatima
- **Records:** 100,000 students
- **Original dataset: 44 columns**
- **Predictor variables used for modeling: 39**
- **Final features after one-hot encoding: 62**
- **Target variable:** `pass_status` with classes Pass / Fail
- **Pass:** 77.389%
- **Fail:** 22.611%

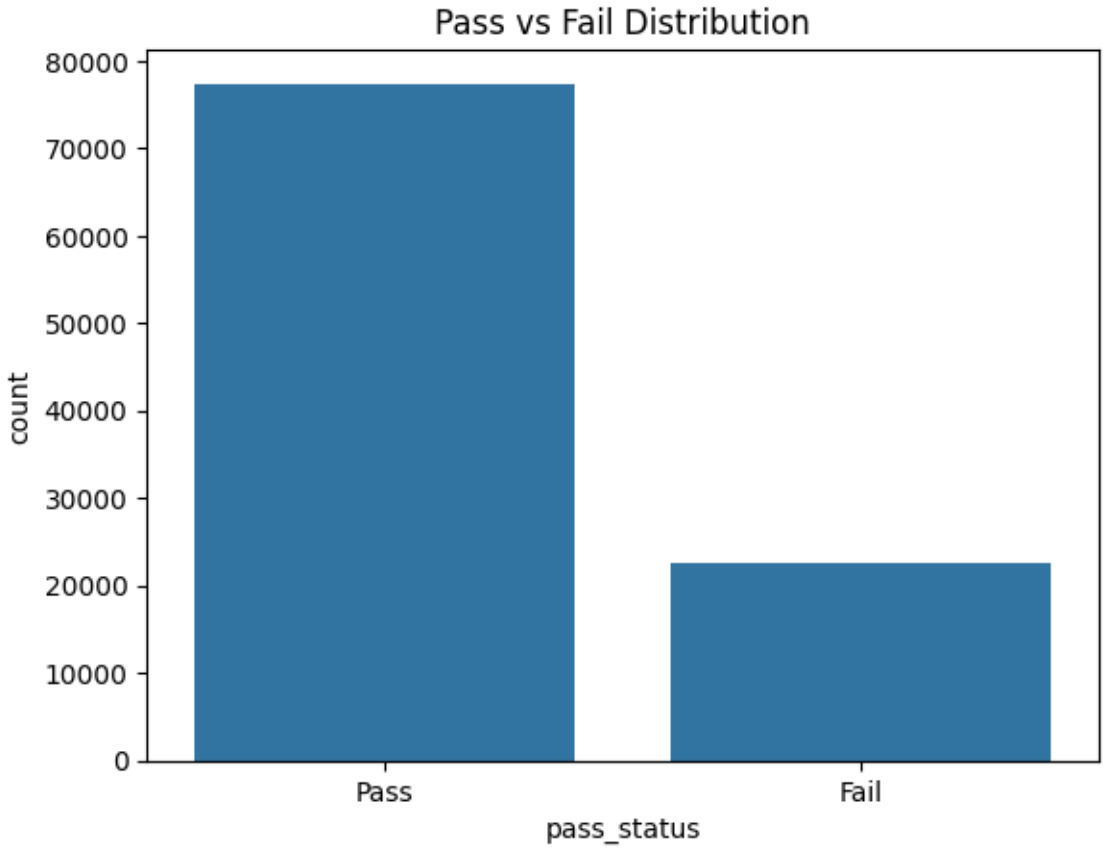
Exploratory Data Analysis

Data Quality

The initial analysis showed that the dataset contains no duplicates. Missing values are present in several variables, particularly `attendance_percentage` (9.86%), `time_management_score` (9.61%), `notes_quality` (8.41%), `previous_gpa` (7.79%), `sleep_quality` (6.99%), and `parent_education` (6.53%). These missing values were considered during the preprocessing stage.

Target Variable

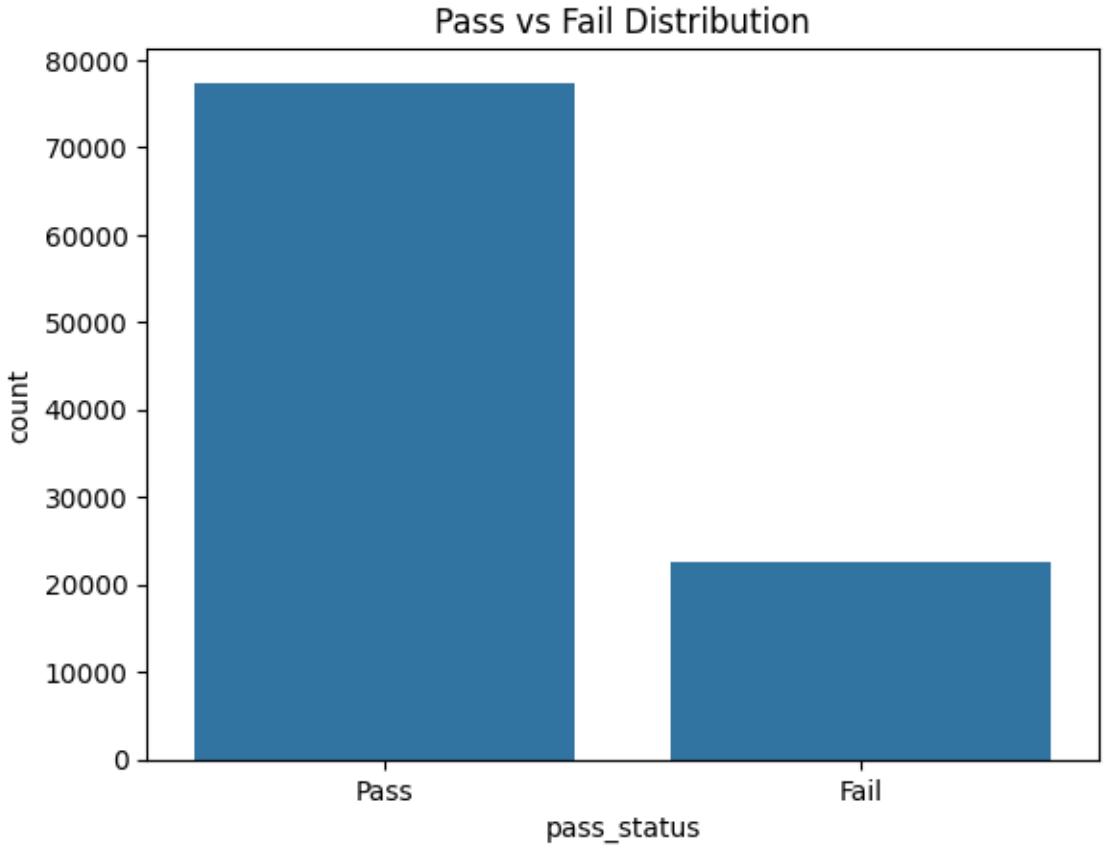
The target distribution shows that **77.39%** of students passed while **22.61%** failed makint the dataset slightly imbalanced toward the passing class. This is important for the modeling stage because accuracy alone may not adequately reflect the model's ability to identify students who are at risk of failure.



Academic Performance Factors

The EDA examined previous academic performance, including `previous_exam_score` and `previous_gpa`, as potential indicators of examination success. Attendance and study-related variables were also explored to investigate whether students' academic engagement and preparation are associated with their outcomes.

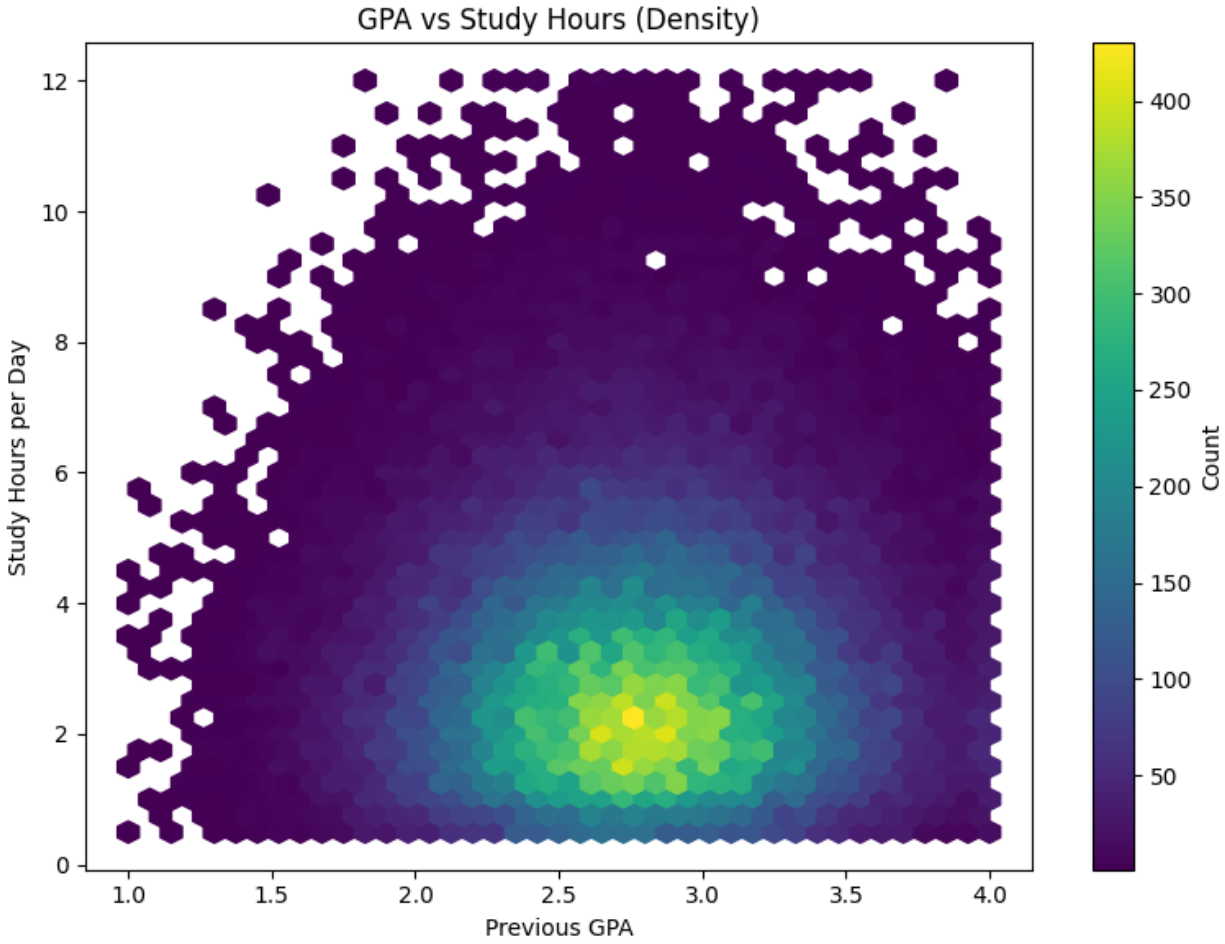
Figure 2. Previous Exam Score vs. Previous GPA Score



Study and Behavioral Factors

Study behavior was investigated through variables such as `study_hours_per_day`, `self_study_hours`, `exam_preparation_days`, and `study_method`. These variables provide insight into how students' preparation habits may relate to their examination outcomes.

Figure 3. Study Hours and Previous GPA



Overall EDA Findings

The EDA establishes several important characteristics of the dataset:

- The target variable is moderately imbalanced, with approximately **23%** of students failing.
- Several variables contain missing observations and require appropriate preprocessing.
- Previous academic performance and study behavior are important variables to investigate further.
- Student wellbeing and behavioral characteristics provide additional dimensions for understanding academic risk.
- The EDA provides the basis for identifying relevant predictors and developing a model capable of identifying students at risk of failure before the examination period.

5. Conclusion, Recommendations & Future Work

5.1 Main Findings

The analysis and modeling produced several important findings. **Previous academic performance** was the strongest predictor of future exam success. **Previous exam score** and **previous GPA** were noticeably more important than the other variables. This indicates that students' previous academic performance provides a strong indication of their potential exam outcomes.

Preparation and study behavior were also important. **Exam preparation days, study hours per day, self-study hours, and the number of practice tests completed** were among the most important predictors. This suggests that students' preparation patterns mattered.

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	79.5%	53.14%	78.885%	63.5%
Ensemble (RF + XGB)	**84.645%**	**66.13%**	65.59%	**65.86%**
Tuned Logistic Regression	79.64%	53.22%	**82.15%**	64.59%
Calibrated Logistic Regression	84.6%	**71.18%**	54.29%	61.6%

Among the evaluated models, Our chosen model was the **Tuned Logistic Regression** mainly for it's high recall on fail class The model also achieved an **ROC-AUC of 89.00%**, indicating good ability to distinguish between students likely to pass and fail.

Overall, the results demonstrate that machine learning can be used to identify patterns associated with student failure and provide an opportunity for **early intervention before the examination period**.

5.2 Business Recommendations

Based on the findings, educational institutions could implement an early warning student risk system that uses academic and behavioral information to identify students who may require additional support.

1. Implement an early risk identification system

Students could be assigned risk levels based on their predicted probability of failing, ex:

- **Low Risk**: normal academic support
- **Medium Risk:** additional monitoring and academic guidance
- **High Risk:** immediate intervention and personalized support

This would allow institutions to prioritize their resources toward students who need them most.

2. Focus on students with weak previous academic performance

Since previous exam score and previous GPA were the strongest predictors, students with consistently weak academic performance should be identified early and offered targeted academic support, such as tutoring, revision sessions, or additional instructor guidance.

3. Encourage structured exam preparation

Preparation days, study hours, self-study hours, and practice tests were among the important predictors. Institutions could therefore encourage students to begin preparing earlier and provide structured revision plans and practice examinations.

4. Monitor attendance and assignment completion

Attendance and assignment completion can serve as additional indicators of academic engagement. Students showing declining attendance or incomplete assignments could be flagged for early follow-up before their performance deteriorates further.

5. Include student wellbeing in intervention programs

Sleep, stress, and exam anxiety also appeared among the model's important predictors. Academic interventions should therefore not focus exclusively on studying. Universities and schools could provide time-management guidance, stress-management resources, and academic counseling.

5.3 Limitations & Future Work

Despite the strong predictive performance, several limitations should be considered.

Data limitations:
The dataset is based on a single structured dataset and may not represent all educational environments or student populations. Some variables also contain missing values, which may affect the analysis and model predictions.

Predictive rather than causal analysis:
Feature importance identifies variables that are useful for prediction, but it does not establish that changing a particular variable will directly cause a student to pass or fail.

Model interpretability:
Although ensemble tree-based models achieved strong predictive performance, their decisions are less straightforward to interpret than simpler models. Further explainability techniques could be used to help educators understand individual predictions.

Future work could include collecting real time student data throughout the semester, incorporating additional behavioral and academic indicators, and testing the system on data from different institutions or academic years.

Finally, future versions could move beyond a simple Pass/Fail prediction and estimate a **student's probability of failure and corresponding risk level**, allowing institutions to prioritize interventions according to the severity of the predicted risk.